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Research Paper

Measurement of public acceptance of TDM policies using public policy acceptance (PPA) and value belief norm (VBN) combination approach

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ABSTRACT

One strategy to overcome the problem of imbalance between the demand and supply of road transportation that causes congestion is to apply transport demand management (TDM). TDM is a series of transportation policies aimed at achieving sustainable transportation by reducing the use of private vehicles and prioritizing public transport and/or non-motorized vehicles. The level of public acceptance of TDM largely determines the success of TDM implementation. Through the value belief norm (VBN) theory approach, it can be seen that public acceptance of TDM policies will be influenced by how high the norm of community partiality towards the environment. The level of public acceptance of a TDM regulation can also be measured by the public policy acceptance (PPA) model. The results show that the acceptance of TDM strategy implementation was significantly influenced by the pro-environment attitude of the community. The PPA model result showed that people tend to be skeptical of the implementation of TDM policies.

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1. Introduction

Discussions about sustainable transportation often revolve around issues of citizens' behavior. Extensive car use with nonrenewable fuels is the main problem of road transportation as the imbalance in demand growth (movement using private vehicles) with supply growth (road transportation network infrastructure) (Kresnanto, 2019; Carran-Fletcher et al., 2020) rapidly increases environmental problems (e.g., congestion and air pollution) (Zhang and Batterman, 2013; Jereb et al., 2018; Li et al., 2020; Rocha Filho et al., 2020; Gunawan, 2021). Concern about environmental issues brings more initiatives about pro-environmental behavior in transportation (Mikiki and Papaioannou, 2012), which addresses sustainable transportation. Travel behavior set with boundaries is believed to be a strategic way to make behavior change movement (Eriksson, Garvill and Nordlund, 2006). Transport demand management (TDM) is one of the public policies in transportation embedded in the comprehending way to trigger citizen's changing behavior. Through various schemes (e.g., tax vehicle, fuel tax, cordon pricing, road pricing, parking management, and working hours setting), TDM offers alternative strategies to reduce transportation problems (Eriksson et al., 2006; Bao et al., 2020; Kresnanto and Putri, 2023).

Meanwhile, the prior research highlights the acceptability of TDM measures as a hierarchical set of beliefs (Eriksson et al., 2006). As an early model for the acceptability of TDM measures, this research depicted fundamental beliefs to develop the

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public policy acceptance (PPA) approach (Loukopoulos, 2005). In urban mobility, public policies and individual concern about the environment and sustainability play essential roles in developing pro-environmental behavior in transportation (Steg and Vlek, 1997; Steg, 2007). In framing the interrelation between changing behavior in urban transportation, Eriksson proposed the models which joined the value belief norm (VBN) theory (Stern et al., 1999) and PPA approach (Oxley, 2010; Zvěřinová, et al., 2013), which specified willingness to reduce car use (WR) (Eriksson, Garvill and Nordlund, 2006; Gärling and Schuitema, 2007).

According to Eriksson's model, personal norms (PNs) represent personal value as the principal factor influencing the willingness to reduce cars. Erricson also stated that problem awareness directly influences PNs in the context of pro-environmental behavior. Addressing the acceptability of TDM, Eriksson connects the VBN theory with the PPA approach, which consists of freedom, own reduction, effectiveness, and fairness as an antecedence to influence TDM acceptability as the product of public policies. Eriksson's model is a valuable conceptual background that enlarges the idea of comprehending the pro-environmental behavior model, specifically to propose the relationship between VBN and TDM (as the product of the PPA approach), but not sufficient to explain how the result of TDM measurement impacts on intention or switching to the actual behavior. One of the well-known behavioral theories in environmental psychology (Abrahamse, 2019) is the theory of planned behavior (TPB) (Ajzen, 1991). The TPB stated that the behavior is directly determined by behavioral intention as personal motivation, attitudes towards the behavior, subjective norms, and perceived behavioral control (Ajzen, 1991). TPB is the best framework to clarify the sustained behavior change as a significant challenge (Zavareh et al., 2020), which has to be solved with the comprehending approach depicted from VBN theory as a PN and personal acceptance (PA) of TDM as a public policies product and behavioral theory, which can explain how each concept is interrelated. The study examines the relationship between factors in PNs to influence the acceptability of TDM measures, intention, and behavior. This study expands Eriksson's model by including the intention and (actual) behavior.

2. Literature review

2.1. TDM as public policy for transportation problem solution

Behavioral changes of individual car users are believed to be a vital strategy for conducting sustainable transportation (Eriksson, et al., 2006; Steg, 2007). The WR has been a factor in the VBN model to measure TDM acceptance and reflects the public policy approach in transportation (Loukopoulos, 2005; Eriksson, et al., 2006). TDM strategy is divided into two policy strategies: push and pull (Steg, 2007; Kresnanto and Putri, 2023). Pull policies encourage the use of non-car modes by making them attractive to car users. These policies include transit-oriented development, road reclamation, and rapid bus transit development. In contrast, a push policy discourages car use by making it less attractive, including toll roads, parking fees, and guard fees (Habibian and Kermanshah, 2011). Another prior research divided TDM strategies into noncoercive (pull) and coercive (push), but in political or public policy perspectives, coercive and noncoercive have to be combined (Gärling and Schuitema, 2007) for the best result.

Then, the two major groups of strategies are divided into implementing strategies, as shown in Table 1.

The success of TDM can be measured by single occupant vehicle (SOV) trips reduced and vehicle mile travelled (VMT) reduced as key measured and fuel saved, emissions reduced, cost per vehicle trip reduced, and cost per VMT reduced as supporting measured (Thompson and Suter, 2012). Moreover, the success story of implementing TDM occurred in several managed to reduce SOV from 74.3% to 61.8% during the period 1993–2014 with the commute trip reduction (CTR) program (Bellevue Transportation Commission, 2015); London applies SOV pricing which is by charging a fee on vehicles that only have one passenger (driver only) if entering the city. This strategy can increase public transport use and reduce accidents and pollution (Nelson, 2008); in Milan in 2015, an area pricing strategy reduced traffic by 31.1% on area pricing and 0.4% across the city in a year. This strategy can also reduce CO₂ contamination by up to 35% (Beria, 2016). Furthermore, with 30 years of road pricing experience, Singapore can reduce traffic volume by 10–30% with the electronic road pricing (ERP) scheme (Chin, 2005). ERP in Singapore can increase the use of public transport from 58% in 2008 to 63% in 2012. In addition, ERP can reduced CO₂ pollution by up to 103 kilotons over the last ten years (Institute for Transportation and Development Policy, 2016). Jordan (Amman) was implementing simulation using synchro transportation modeling of 19 TDM strategies coupled with implementing transportation system management (TSM) can reduce delays at intersections and reduce fuel consumption significantly (Jrew et al., 2019).

Furthermore, the results of an inventory of the success of TDM in several cities in the world, based on the type and type of strategy, can be seen in Table 2.

2.2. VBN

VPN was first developed by Stern et al. (Stern et al., 1999), who explained the influence of human values and behavior on pro-environmental orientation. The VBN theory of environmentalism postulates that values influence pro-environmental behavior via pro-environmental beliefs and PNs (Hiratsuka, et al., 2018). VBN is a framework for carrying out a normative investigation of a person and whether his attitudes and behavior are pro-environmental. VBN theory of pro-environmental behavior posited that pro-environmental behaviors are the result of the activation of the personal moral

Table 1

TDM strategy based on literature review.

Policy strategy	General strategy	Operational strategy
Push strategy	Controlling motorized ownership	Vehicle ownership tax scheme
	Controlling motorized vehicle use	Vehicle ownership restrictions Taxes related to vehicle use Transport infrastructure taxes and access restrictions (access management)
Pull strategy	Encouraging non-motorized use	Subsidies for public transport/transit Pedestrian infrastructure improvement (walking) Cycling infrastructure improvement (cycling)
	Encouraging public transport use	Transit services improvement
		Progressive vehicle tax Vehicle sales tax Vehicle ownership permit Fuel tax Road pricing Cordon/area pricing Congestion pricing Plat based restriction Parking management Car-free day Working hours setting Transit operational subsidies Fare subsidies Sidewalk/crosswalk Pedestrian area Bicycle lanes Bicycle parking area Bicycle service points Integrated transit services (schedule, fare) Rapid transit development (e.g., bus lanes and bus priority at the junction) Improve public transport facilities (station, bus stop, etc.)

Sources:(Kresnanto and Putri, 2023).

norms in which activated by beliefs about awareness of consequences, ascribed responsibility to act, and environmental concern about oneself (egoistic) and others (social altruism), and non-human organism (biospheric) (Aziz and Ajuhari, 2014).

In implementing TDM policy, one's VBN to the environment will significantly affect the acceptance of the TDM strategy that the government will carry out. PA and PNs on environmental conditions due to congestion will encourage public interest and participation in the success of sustainable transportation (i.e., TDM) programs. One measure that a person is interested in and desires to participate in sustainable transportation is his willingness to reduce the use of private vehicles (Jakobsson, Fujii and Gärling, 2000; Nordlund and Garvill, 2003). TDM-VBN model can be described in Fig. 1. If the community has good PA and PNs towards the environment, it will be pro to TDM policy by trying to reduce WR.

2.3. PPA

In the application of regulations or policies, especially public policy, the most crucial reason that society will accept or reject is related to social-psychological factors and policy-specific beliefs about fairness and infringement on freedom (IF) (Oxley, 2010; Zvěřinová, et al., 2013). So it can be said that public acceptance of a public policy depends very much on its views on policy, whether it will provide justice and freedom.

If TDM is seen as a form of public policy, the framework for measuring public acceptance of TDM can be approached with PPA (see Fig. 2). Measurement of TDM success in Indonesia seen from the outcome, in general, has never been done, but descriptive measurement of the level of public acceptance of how many TDM strategies have been carried out in Jakarta. This research revealed that the majority of road users are willing to participate in TDM (Rachmat and Pitaloka, 2010). Road pricing policies in some countries have faced significant public opposition due to two reasons, namely (1) it is considered unfavourable to low-income people and tourists, and (2) it is an unpopular policy related to politics because it is considered as a new type of tax (Qin et al., 2022).

3. Materials and methods

3.1. Proposed structural equation model (SEM)-based TDM acceptance framework

The analysis used the SEM method. SEM allows direct analysis among several dependent (exogenous) and independent (endogenous) variables (Hair et al., 2010).

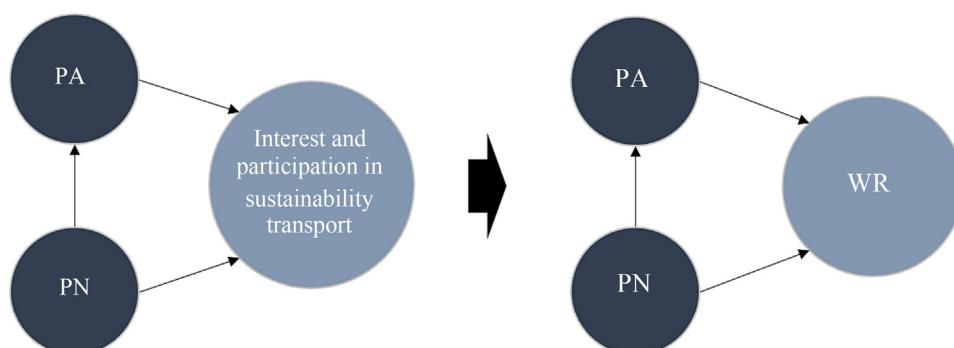
Fig. 3(a) shows the structural model framework with SEM on the proposed public acceptance of TDM policies model. This model is a combination of three theoretical approaches. The first theory is VBN, which assumes that TDM acceptance will be influenced by one's partiality to the environment (pro-environment) because TDM is a pro-environment policy (sustainable transportation). A person's partiality to the environment is shown by the endogenous variable WR, which is influenced by two exogenous variants (PN and PA).

Table 2

Experience in TDM Implementation effectiveness from various world cities.

Case study	Type of policy TDM strategy	Operational TDM strategy	Effectiveness measurement	Source
Southern California	Pull strategy	Telecommuting	Vehicle miles traveler (VMT) reduction (91.1%), trips reduction (1.0%), and CO ₂ emission reduction (1.0%) (Cameron, 1991; Harvey and Deakin, 1991)	(Kusumantoro et al., 2009)
Lloyd District, Portland	Pull strategy	Transit, bicycling, car sharing, walking and ridesharing	SOV dropped by 40% and dropped by 4.3 million peak-hour vehicle miles travelled in the last ten years	(EcoDistricts, 2016)
Bellevue Washington	Push strategy	CTR	Reduced SOV from 74.3% to 61.8% during the period 1993–2014	(Bellevue Transportation Commission, 2015)
Milan	Push strategy	Area pricing	Reduced traffic by 31.1% in area pricing and 0.4% citywide in a year. This strategy can also reduce CO ₂ contamination by up to 35%	(Beria, 2016)
Singapore	Push strategy	ERP	Reduce traffic volume by 10–30%	(Chin, 2005)
USA	Pull and push strategy	Combination of monetary incentive, public transport, transportation service	Increase vehicle travel reduction (VTR) by 10–50% depending on the type of combination of wisdom	(United States. Department of Transportation, 2020)
Auckland	Push strategy	Parking management	Reduce vehicle trips by 8–18% and reduce the level of drive-alone by around 2–5% (Auckland Regional Council, 2000)	(Kusumantoro et al., 2009)
Amsterdam	Pull strategy	Cycling, public transport, and enacting parking policies	63% of Amsterdam residents use their bikes daily and an estimated 61% of trips are undertaken by walking or cycling	(Carran-Fletcher et al., 2020)
London	Pull strategy	Improving public transport	Reduced private vehicle usage from 49% to 36% in 20 years	(Carran-Fletcher et al., 2020)
Seattle	Pull and push strategy	Interventions based on integrated transport and land use planning, CTR policies, and investment in public transport	Dropped drive-alone rates from 35% to 25% (2010–2017)	(Carran-Fletcher et al., 2020)

The second theory, PPA theory, views that people's partiality towards a public policy (TDM endogenous variable, the acceptance of TDM strategy (AC)) will be influenced by exogenous variables of fairness norms and belief in the absence of IF. The third theory, The theory of planned behaviour (Ajzen, 1991), is to complete how this process in the VBN and PPA theory becomes an intention and realize (actual) behavior. So, accepting a public policy, TDM will affect intentions, leading to actual behaviour towards TDM. In this paper, the TDM strategy chosen to build a structural model was a TDM strategy that has been widely implemented, e.g., road pricing (Chin, 2005), plate license restriction (Supriana et al., 2020), and working hour setting (Giuliano and Golob, 1990).

**Fig. 1.** VBN model for sustainable transportation (TDM policy).

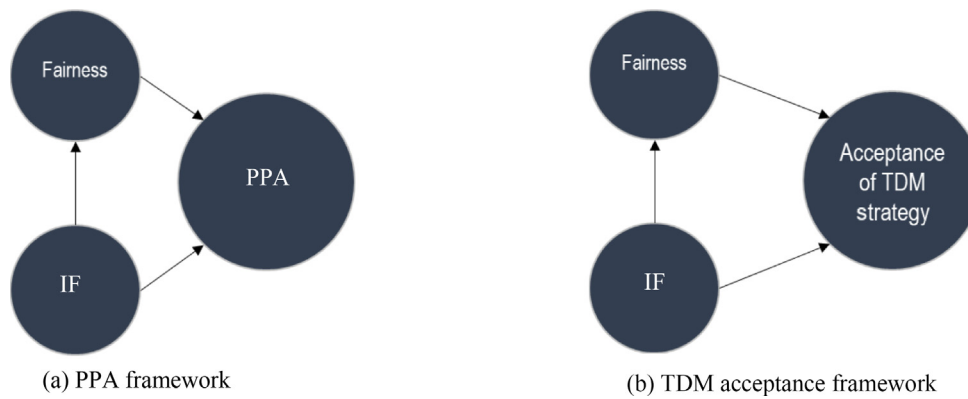


Fig. 2. PPA vs TDM acceptance framework.

Based on the research objectives, this proposed model extends and modifies Eriksson's model. The comparison between Eriksson's and proposed models is shown in Fig. 3. Eriksson's model is connected to the willingness to reduce with all the acceptability of the TDM measure's antecedent. So, the relationship between willingness to reduce and acceptability of TDM is measured through variables such as freedom, own reduction, effectiveness, and fairness (see Fig. 3(a)). Meanwhile, the proposed model does not connect the WR through the variables of freedom and fairness but directly to the TDM strategy's acceptability and interrelation with intention and behavior (see Fig. 3(b)).

3.2. Study location, procedure, and respondents

This research was conducted by a survey to measure the indicators of the TDM acceptance model on respondents in the Special Region of Yogyakarta (DIY). DIY was selected as the research location due to its relatively high personal vehicle growth rate of over 10% annually (Kresnanto, 2019). The questionnaire was distributed randomly in July 2023 in the DIY Province region, with the respondent residential dispersion depicted in Fig. 4.

Respondents were people who travelled every day to the Yogyakarta City area. With a DIY population of 4 021 816 people, the minimum sample size based on the Slovin formula (Woods, 2022) with a margin of error of 5% is about 400 respondents. The number of respondents obtained during the survey was 451, with the characteristics in Table 3.

3.3. Questionnaire

A questionnaire was designed on the TDM acceptance structural model (acceptance of road pricing model, acceptance of plate-based restriction model, and acceptance of working hour setting model) with the PPA and VBN combination approach as in Table 4. All question items were modified from previous research (Eriksson, et al., 2006; Jou et al., 2010; Gibson and Carnovale, 2015; Hsieh, 2022). Measurement of answers to the questionnaire used the Likert scale 1–7 levels (1 = strongly disagree, 2 = more disagree, 3 = disagree, 4 = undecided, 5 = agree, 6 = more agree, 7 = strongly agree). These 7 Likert scales provided a much broader picture of respondent's opinions (Bernstein, 2005). Moreover, the Likert 7 scale was chosen to get responses with great accuracy. The rating scales that yielded the least reliable scores turned out to be those with the fewest response categories on several indices of reliability, validity, and discriminating power, the two-point, three-point, and four-point scales performed relatively poorly, and indices were significantly higher for scales with more response categories, up to about 7 (Preston and Colman, 2000). 7-point scales strongly correlated with *t*-test results (Lewis, 1993). A 7-point Likert item is more likely to reflect a respondent's proper subjective evaluation of a usability questionnaire than a 5-point item scale (Finstad, 2010).

4. Result and discussions

The three models built can be considered suitable or appropriate judging from the threshold value of standardized root mean residual (SRMR) of < 0.10 or less (Henseler et al., 2014), and the value of acceptance score normal fit index (NFI) is > 0.80 (Reinard, 2006). Based on Table 5, the value of estimated model on plate license restriction is 0.075 (< 0.10) and the value of estimated model on working hour setting is 0.081 (< 0.10), it means that both model indices an appropriate model fit. However, the model that was considered the best is the acceptance of working hour setting model with better SRMR and NFI values compared to other models.

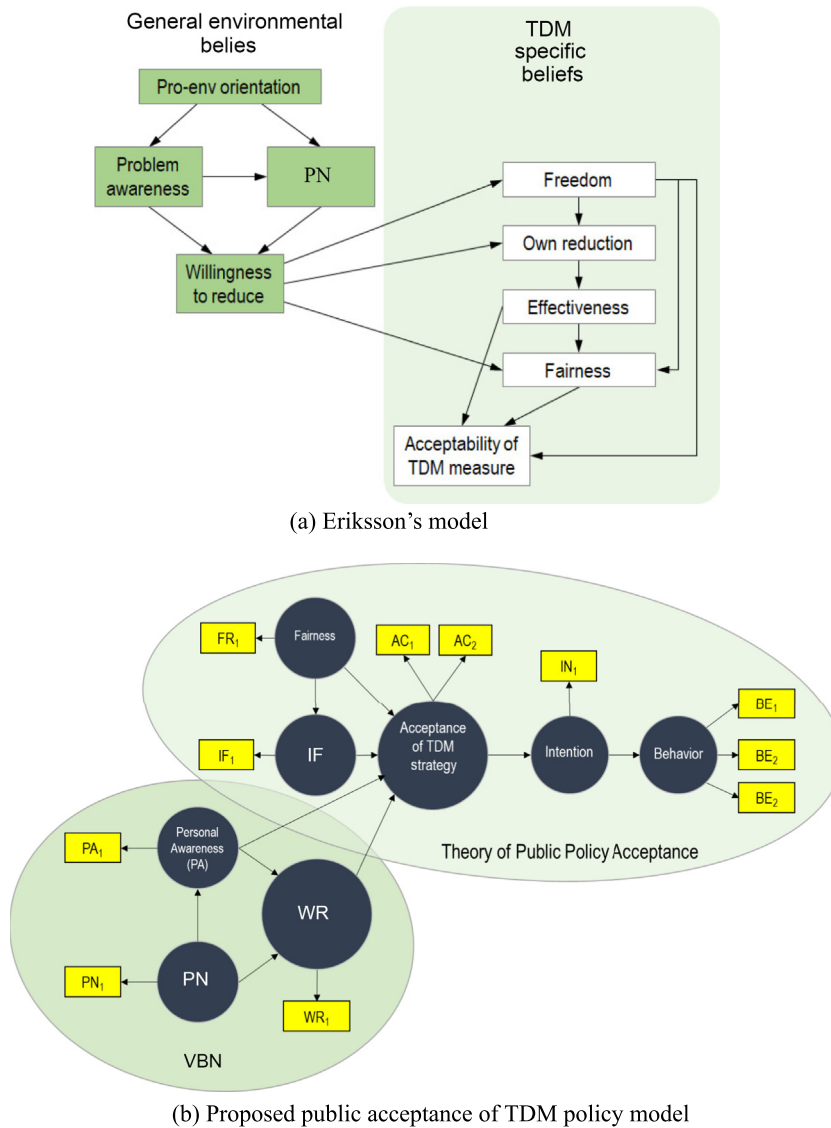


Fig. 3. Proposed TDM acceptance structural model.

4.1. Descriptive analysis (VBN and PPA)

Table 6 shows the average results and standard deviations (SDs) of assessment by respondents against VBN and PPA variables. In VBN, problem awareness respondents showed a good response (mean > 5) related to awareness about the bad consequences of motor vehicle pollution. Likewise with the PN, with a mean close to 5. However, a good enough PA and PN did not guarantee that respondents will have an interest and will participate in sustainable transportation, as evidenced by the mean in WR smaller than PA and PN.

Meanwhile, the other variables had almost the same average and are at a sufficient level.

Fig. 5 depicts the respondents' response trends' results due to hierarchical beliefs from problem awareness, PNs, and WR. The bottom of the bar chart (PA) shows dominance in strong agreement (green area), which diminishes as it shifts upward to PN, the green area gets narrower, and the top chart (WR) shows that the green area is increasingly narrow. It means that problem awareness is not necessarily followed by increasing PNs and WR but by decreasing. Based on the data, it can be seen that problem awareness in the community should be increased extensively to result in better trend in pro-environmental behavior.

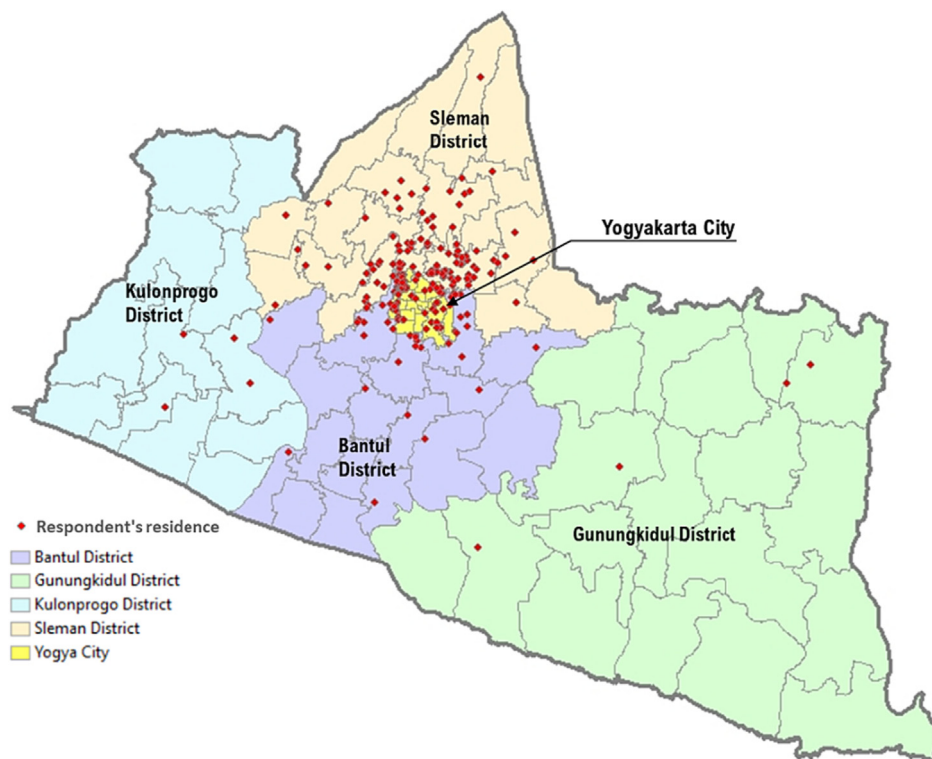


Fig. 4. Spatial distribution of respondents' residence.

Table 3
Characteristics of respondents.

Characteristics	Number of respondents	Percent (N = 419)
Age		–
15–20	59	14.08
21–30	245	58.47%
31–40	55	13.13%
41–50	48	11.46%
51–60	8	1.91%
> 60	4	0.95%
Gender		
Male	251	59.90%
Female	168	40.10%
Education level		
Elementary school	3	0.72%
Junior high school	19	4.53%
Senior high school	251	59.90%
Diploma	17	4.06%
Bachelor's degree	115	27.45%
Master's degree	11	2.63%
Doctoral degree	3	0.72%
Income per month (Rupiah – Rp)		
< Rp2 500 000	277	66.11%
Rp2 600 000 – Rp5 000 000	85	20.29%
Rp5 100 000 – Rp7 500 000	26	6.21%
Rp7 600 000 – Rp10 000 000	15	3.58%
Rp10 100 000 – Rp12 500 000	7	1.67%
Rp15 100 000 – Rp20 000 000	5	1.19%
> Rp20 000 000	4	0.95%
Main transport mean		
Walking	14	3.34%
Cycling	6	1.43%

(continued on next page)

Table 3 (continued)

Characteristics	Number of respondents	Percent (N = 419)
Motorbike	348	83.05%
Car	43	10.26%
Transit	8	1.91%
Purpose of daily travel		
Working	259	61.81%
Shopping	2	0.48%
Studying for bachelor's degree	49	11.69%
Studying in junior/senior high school	66	15.75%
Other	43	10.26%

Table 4

Questionnaire for indicator measurement.

Theory	Latent variables	Observed indicators	The nature of the question
VBN Pro environment orientation	Problem awareness	(PA ₁) I believe that pollution from motor vehicles is very bad for people and the environment? (Eriksson, et al., 2006)	Positive
	PN	(PN ₁) I feel morally responsible to reduce the use of private vehicles in order to reduce the negative impact of pollution and fuel wastage? (Eriksson, et al., 2006)	Positive
	WR	(WR ₁) I agree/am willing to reduce the use of private vehicles in order to reduce the negative impact of pollution and fuel waste? (Eriksson et al., 2006)	Positive
PPA	Fairness (FR.)	(FR ₁) In my opinion, the TDM (road pricing/plate number restriction/working hour setting) policy is unfair to me and other drivers (Jou et al., 2010)	Negative
	IF	(IF ₁) In my opinion, The implementation of TDM (such as road pricing, plate number restriction, working hour setting) is a violation of freedom in choosing routes due to the increase in travel costs (Jou et al., 2010)	Negative
	Acceptance	(AC ₁) In my opinion, TDM policies (such as: road pricing, plate number restriction, working hour setting) are positive policies to overcome congestion (Jou et al., 2010)	Positive
	Intention	(AC ₂) If you vote in a referendum, it is likely will I receive a TDM policy? (Jou et al., 2010)	Positive
	Behavior	(IN ₁) I agree with the implementation of TDM (such as road pricing, plate number restriction, and working hour setting) policy to reduce congestion (Hsieh, 2022)	Positive
		Acceptance of road pricing model (BRP ₁) I will still go through the road with the application road pricing and not looking for alternatives (Gibson and Carnovale, 2015)	Positive
		(BRP ₂) I will find an alternative way to avoid road pricing (Gibson and Carnovale, 2015)	Negative
		(BRP ₃) I will change the travel time so as not to be subject to the road pricing policy (Hsieh, 2022)	Negative
		Acceptance of plate license restriction model (BPL ₁) I will use public transportation if my vehicle plate is subject to the odd-even policy	Positive
		(BPL ₂) I will use another vehicle whose license plate number matches odd/even	Negative
		Acceptance of working hour setting model (BWS ₁) I will go to work/school according to the set hours	Positive

Table 5

Model Fit.

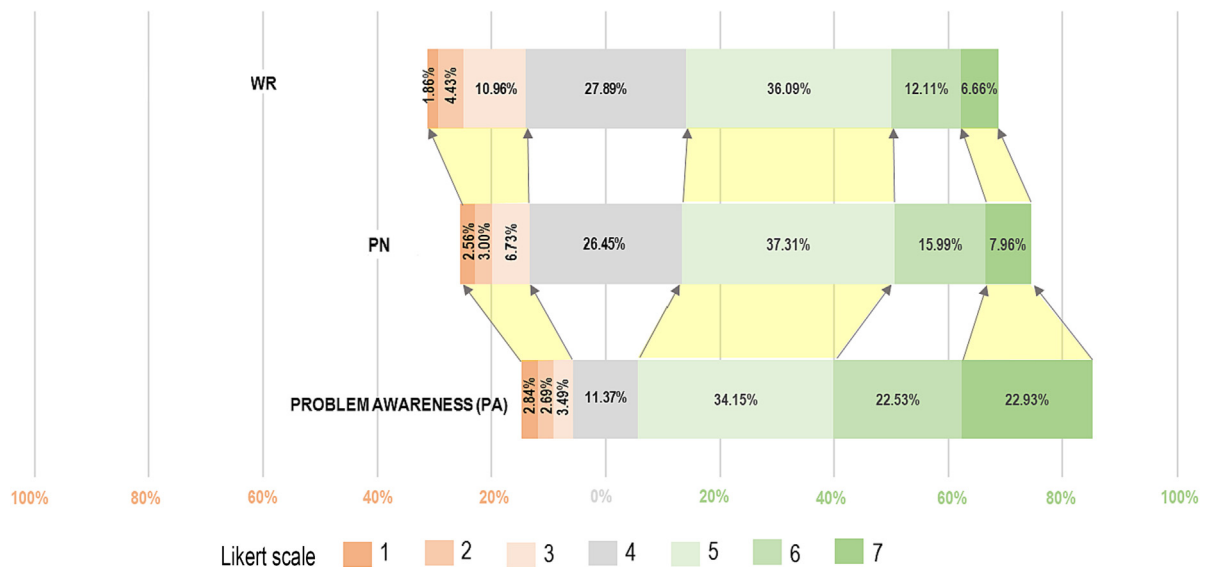
	Acceptance of road pricing model		Acceptance of plate license restriction model		Acceptance of working hour setting model	
	Saturated model	Estimated model	Saturated model	Estimated model	Saturated model	Estimated model
SRMR	0.092	0.122	0.056	0.075	0.030	0.081
NFI	0.706	0.657	0.789	0.760	0.891	0.843

Table 6

Mean and SD for VBN and PPA variables.

		Acceptance of road pricing model		Acceptance of plate license restriction model		Acceptance of working hour Setting Model	
		Mean	SD	Mean	SD	Mean	SD
VBN	Problem Awareness (PA)	5.304	1.356	5.457	1.226	5.401	1.293
	PN	4.922	1.337	4.889	1.203	4.878	1.221
	WR	4.698	1.264	4.654	1.226	4.685	1.236
PPA	Fairness (FR.)	4.149	1.183	4.326	1.261	4.126	1.112
	IF	4.432	1.212	4.308	1.116	4.215	1.055
	AC1	4.621	1.272	4.472	1.188	4.494	1.095
	AC2	4.412	1.282	4.237	1.138	4.397	1.068
	Intention	4.647	1.427	4.375	1.253	4.497	1.162
	BRP1	4.304	1.384				
	BRP2	4.297	1.375				
	BRP3	4.601	1.397				
	BPL1			4.408	1.214		
	BPL2			4.169	1.207		
	BWS1					4.641	1.142

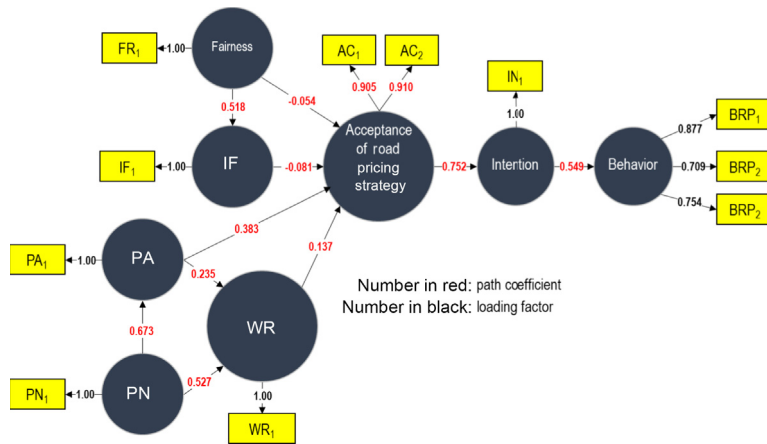
Scales 1–7 (1 = strongly disagree, 2 = more disagree, 3 = disagree, 4 = undecided, 5 = agree, 6 = more agree, 7 = strongly agree).

**Fig. 5.** Hierarchical beliefs from problem awareness, PNs, and WR.

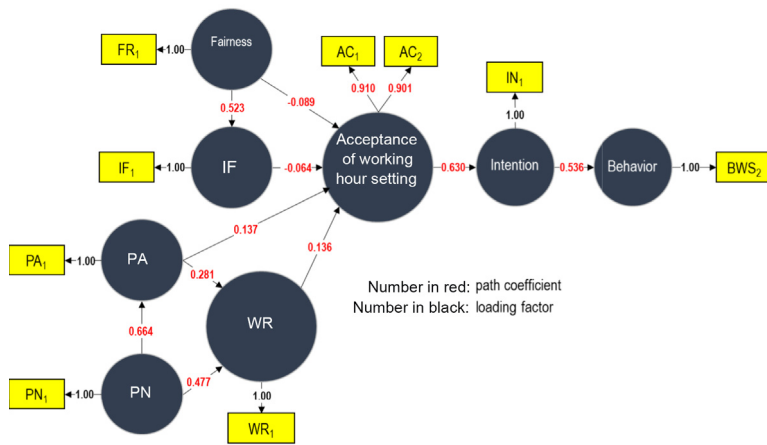
4.2. Path analysis

Path coefficients are helpful in showing the direction of the relationship between variables in SEM, which ranges from -1 to 1 . The closer to 1 , the stronger the positive relationship, and vice versa; the closer to -1 , the stronger the negative relationship, and vice versa. Otherwise, a value close to 0 means that the two variables have no relationship (Ayer et al., 2017). The results of the SEM path analysis on three TDM strategies (see Fig. 6) show that most endogenous variables positively affected the exogenous variables, except for PPA. In the case of implementing TDM policy, it turned out that fairness and IF considerations can be ruled out, while PA and PN communities are the key to the success of implementing TDM policy. Acceptance attitudes towards pro-environment policies are further strongly influenced by gender, socioeconomic and education levels (Karpudewan, 2019), and especially knowledge (Liobikien and Poškus, 2019).

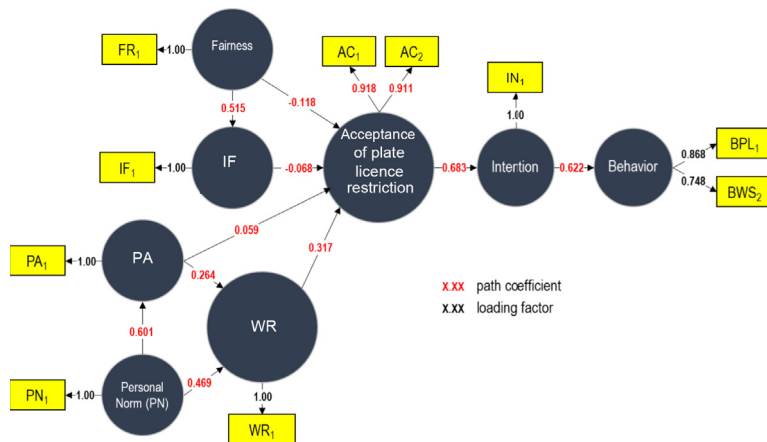
In VBN, the correlation of endogenous variables to exogenous varies; this is in line with research conducted by Ghazali et al. (2019) that the socioeconomic conditions of the community strongly influenced the influence of PA and PN on WR. Nevertheless, PN significantly influenced PA in all TDM strategy implementations.



(a) Loading factor and path coefficient road pricing acceptance



(b) Loading factor and path coefficient working hour setting acceptance



(c) Loading factors and path coefficients for VBN and PPA in TDM

Fig. 6. Loading factors and path coefficients for VBN and PPA in TDM.

The path coefficient in PPA (in all models) showed that perceptions of fairness and IF did not significantly affect AC or had a very small/near-zero negative influence (see Fig. 6). This result was also supported by the results of bootstrapping analysis on the structural model which shows that there is no correlation between the variables PPA (fairness and IF) and acceptance

Table 7
Bootstrapping Analysis Results.

	Road Pricing		Plate License Restriction		Working Hour Setting	
	T-Sta	P-Values	T-Sta	P-Values	T-Sta	P-Values
Acceptance of TDM Strategy -> INTENTION	26.445	0	17.774	0	13.846	0
Fairness -> Acceptance of TDM Strategy	0.925	0.355	1.859	0.064	1.202	0.229
Fairness -> IF	11.642	0	8.993	0	9.818	0
INTENTION -> BEHAVIOR	14.868	0	14.371	0	9.214	0
IF -> Acceptance of TDM Strategy	1.414	0.158	1.025	0.306	0.788	0.431
PN -> Problem Awareness	19.638	0	11.694	0	17.409	0
PN -> WR	7.695	0	6.993	0	8.304	0
Problem Awareness -> Acceptance of TDM Strategy	6.69	0	1.02	0.308	2.039	0.042
Problem Awareness -> WR	3.975	0	4.146	0	4.919	0
WR -> Acceptance of TDM Strategy	2.284	0.023	5.653	0	2.307	0.021

of TDM strategy, indicated by p values > 0.05 and t -stat > 1.96 . This condition can be considered as a condition that the community has a skeptical attitude toward a policy that will be implemented.

The plate license restriction model also found that a variable that did not significantly affect acceptance of TDM strategy, was problem awareness, with a p -value of < 0.05 .

The results showed that perceptions of fairness and IF did not affect PPA (see Table 7). This finding might occur due to the public's scepticism or indifference towards applying public policy. In the other hand, based on VBN theory, p -value score in all paths on model working hour setting and road pricing model are statistically significant to shaping (actual) behavior.

5. Conclusions

The results showed that public acceptance of the TDM strategy (road pricing, license plate restriction, and working hour setting) was influenced by pro-environment attitudes (WR). This pro-environment attitude was an accumulation of the influence of PNs and PA of the community in responding to environmental conditions due to road transportation problems. However, based on PPA modelling, the DIY community still tends to be sceptical of implementing TDM policies. This was indicated by the path coefficient value between policy social-psychological factors (fairness and IF) acceptance of TDM strategy, which is close to zero. The highlighted relationship between VBN and PPA enriches the research in TDM and enables another scholar to examine the phenomena in contextual ways.

TDM strategy can be effective if it does not only implement one strategy but must be a combination of several strategies. Therefore, future research should focus on measuring the acceptance of combined TDM strategies, especially the combination of push and pull that is applied together.

CRedit authorship contribution statement

Nindyo Cahyo Kresnanto: Conceptualization, Funding acquisition, Data curation, Writing – original draft, Writing – review & editing, Visualization, Formal analysis, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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